AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I162*

Team Name: *KDVS*

Domain: *ADVANCED MATERIALS,MANUFACTURING AND INDUSTRIAL INNOVATION*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Use Cases
9. Impacts and Benefits
10. Sustainable Development Goals
11. Individual Member Contribution

Problem Statement

How can small manufacturers cut electricity cost without reducing production ?

Problem Understanding

- Small and medium manufacturers, such as foundries, pump and motor makers and workshops, that run heavy electrical machines.
- Machines are scheduled by habit and shift convenience, not by electricity price.
- Machine operators and shift staff, maintenance electricians, accounts staff who handle the bill, industry associations, and the power distribution company that must meet peak demand.

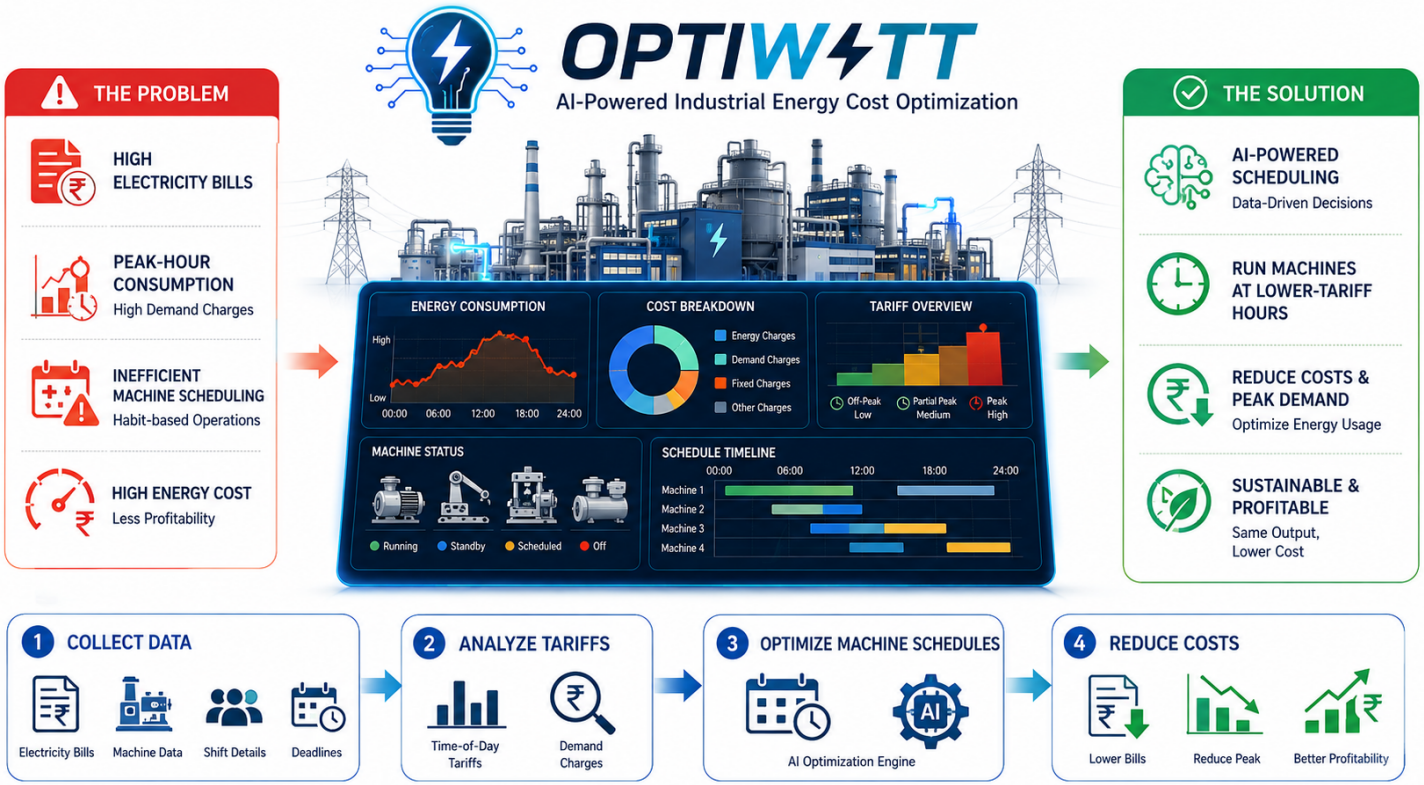
Existing Gaps & Challenges

- Machines are scheduled by habit and shift convenience, not by electricity price.
- Owners see only one total bill a month, so they cannot tell which machines or hours cost the most.
- How can a small manufacturer use data it already has (bills, shift timings, machine list) to schedule heavy machines away from peak hours and lower energy cost without reducing production ?

Our Proposed Solution

- Build a software-based solution that combines electricity-bill analysis, production scheduling, and demand-charge prediction, the tool reads the tariff, shows cost per machine and hour. Instead of simply telling factories to use less electricity, it tells them which machines to run, when to run them, and how to maintain the same production output at a lower electricity cost.
- It reduces electricity costs through smarter machine scheduling and reduced peak-hour usage, helping manufacturers save money while maintaining production and meeting deadlines.
- Lower electricity bills, smarter machine scheduling, and the same production output,without expensive hardware

Technical Approach



[Explain how the system components work together.]

Tech Stack

Frontend: [Add here]

Backend: [Add here]

Database: [Add here]

AI/ML, APIs, or Hardware: [Add if applicable]

Innovation & Uniqueness

- It combines electricity-bill analysis with smart machine scheduling. It recommends the best times to operate machines, avoid peak-demand charges, and meet production deadlines. Unlike expensive energy management systems, it aims to provide an affordable software-based solution using existing data, without requiring costly new hardware.

- It is suited to this problem because it uses existing electricity bills, machine details, and production schedules to recommend cost-effective machine timings. It helps small manufacturers avoid expensive peak hours and demand spikes while meeting production deadlines, without requiring costly hardware or reducing output.

- The key differentiator is combining electricity tariff optimization with production-aware machine scheduling. It recommends when to run machines to reduce electricity costs while maintaining production output and meeting deadlines, without requiring expensive new hardware.

Feasibility & Scalability

- **Feasibility:** Build a software-based prototype using Python, Streamlit, and MILP optimisation to analyse electricity bills and recommend cost-effective machine schedules. Start with simulated data and integrate energy meters later.
- **Scalability:** Use one common platform with reusable machine profiles and industry-specific production constraints. Expand from a few machines to multiple machines, factories, and industries without rebuilding the entire system
- **Risks:** Inaccurate data can lead to unreliable savings estimates. Production delays can be prevented by enforcing deadlines and machine constraints.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Small-Scale Manufacturing Factory — A factory owner uses SmartEB to analyse machine-wise electricity consumption and optimise operating schedules based on tariff rates, reducing electricity costs while maintaining production output and meeting deadlines.

- Use case 2: CNC / Metal Fabrication Workshop — A workshop manager uses SmartEB to schedule flexible machining jobs during lower-tariff periods and reduce simultaneous peak power demand where applicable, without delaying customer orders.

- Use case 3: - Textile Industry — A textile unit uses SmartEB to monitor machine energy consumption, identify unnecessary idle usage, and optimise the scheduling of flexible operations while respecting continuous-process requirements and production targets
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- Benefits to target users: Helps small-scale manufacturers reduce electricity bills, optimise machine schedules, minimise unnecessary energy usage, and maintain production output without requiring expensive hardware initially.
- Social, economic, and environmental impact: Reduces operating costs for small businesses, improves energy efficiency, supports business sustainability, and can lower carbon emissions when electricity consumption is reduced.
- Expected measurable outcomes: Lower electricity costs, reduced avoidable idle-energy consumption, improved machine utilisation, and fewer peak-demand charges where applicable. Savings will be measured by comparing baseline and optimised bills while maintaining equivalent production output.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
7	7.3	Affordable and Clean Energy	supports improved energy efficiency through smarter scheduling.
9	9.4	Industry, Innovation and Infrastructure	Helps smaller industries explore more efficient production operations

Individual Member Contribution

Member	Contribution or responsibility
SASWANTH K	Coordination, problem framing, pitch flow and integration.
KARUN SENTHILKU-MAR	User research, tariff/bill data collection and validation.
DEVESHVAR SURESHBABU RAMYA	Scheduling logic, constraints and prototype testing.
MOHAMMED RAKSHAN I	Interface design, diagrams, documentation and demo preparation.

Yukti ID

IR2026-1051087